



Expernetic (Pvt) Ltd

Talent Assessment Program · Technical Assignment

Data Engineer Intern

Assignment

Assumptions & Decisions Log

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The following log details the core technical assumptions and architectural decisions made throughout the lifecycle of this project. These choices were guided by the need to balance data integrity with the constraints of an analytical environment.

Summary of Key Technical Decisions

ID	Category	Decision	Justification
D01	Architecture	SQLite Analytical Engine	Provides high portability and efficient SQL query support without the overhead of a dedicated cloud server.
D02	Modeling	Star Schema implementation	Separates volatile facts from static dimensions, optimizing performance for BI dashboard aggregations.
D03	ETL	Regex-based price cleaning	Necessary to convert web-scraped strings ("-\$1,200.00") into floats for mathematical analysis.
D04	Geospatial	Haversine distance calc	Needed to transform lat/long coordinates into a continuous "distance-to-center" feature for predictive potential.
D05	Visualization	Embedded reporting	Placing charts alongside interpretation in Section 6 improves

			stakeholder readability vs. an appendix.
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Deep-Dive Justifications

Architectural Decisions

- **Decision (D01 & D02):** I chose to implement a **Star Schema** within an **SQLite database**. While a flat CSV file is sufficient for small scripts, a relational model is the industry standard for Data Engineering. By separating the `Fact_Listings` table from the dimension tables (`Dim_Host`, `Dim_Date`), I ensured that the data model is scalable. If the project were to grow, this structure allows for easy migration to a production data warehouse like BigQuery or Snowflake without re-writing the core logic.
- **Decision (D05):** The decision to integrate all visual figures directly into the findings section (Section 6) rather than an appendix was a deliberate UX (User Experience) choice. It allows a non-technical stakeholder to digest a business trend and the supporting visual proof simultaneously, minimizing cognitive load.

Data Assumptions & Cleaning Rules

- **Assumption (A01 - Imputation):** I assumed that properties with missing `beds` or `bedrooms` information but valid active status should be imputed with a value of 1.

Rationale: Deleting these rows would have removed ~15% of the total dataset, potentially introducing significant selection bias. Imputation allows us to retain the financial and geographic data of these listings.
- **Assumption (A02 - Occupancy Proxy):** I operated under the industry-standard assumption that days marked `available = f` (false) in the calendar logs represent booked nights.

Caveat: I explicitly acknowledge that this may slightly overstate occupancy, as these days could also represent blocked time for host personal use or maintenance.
- **Assumption (A03 - The "Review Inflation" Bias):** I assumed that the star-rating variable is not a reliable indicator of quality due to the observed 85%+ rating inflation.
- *Rationale:* This drove the decision to pivot the analysis toward price and host tenure as more objective proxies for listing success and market value.

Engineering Constraints

- **Decision (D03):** When encountering scraped data, I decided to use `pd.to_numeric` with `errors='coerce'` after regex cleaning.
Outcome: This ensures that any data that could not be cleaned (e.g., truly corrupt rows) was converted to `NaN` and effectively handled rather than causing the entire pipeline to crash.